EE3400 – Electronic Measurements and Instrumentation

Tutorial 2: Sensors, Strain Gauges and Instrumentation Amplifiers

Purpose: This tutorial consolidates the material covered in Lectures 1–3 and follows the style and level of the supplied sample tutorial.

Instructions: Show all working for numerical questions. Use SI units unless otherwise stated. Where an approximation is specified, use the stated relation.

Sensors

1. Define **measurement** and explain why a standard unit is required in measurement.

2. Distinguish between a **sensor/transducer**, an **actuator**, and a **measurand**.

3. A temperature sensor reads 47 °C while the actual temperature is 50 °C. Calculate the percentage error and percentage accuracy.

4. A temperature sensor has a measurement range of −20 °C to 80 °C. Determine: (a) the range, and (b) the span.

5. An LM35 sensor has a sensitivity of 10 mV/°C. What change in output voltage is expected for a temperature increase of 7 °C?

6. A 16-bit ADC is used over a 0–5 V span. Using the resolution relation $Resolution = Span/2^N$, calculate the voltage resolution. Comment briefly on what this means.

7. Explain the difference between **accuracy**, **precision**, and **reproducibility**.

8. What is sensor **bandwidth**? Why is it important when selecting a sensor for a measurement application?

9. Explain **hysteresis** and **non-linearity/non-conformity** as sources of measurement error.

Strain Gauges

10. Define mechanical strain and write its mathematical expression. A 50 mm specimen elongates by 0.025 mm. Calculate the strain in με.

11. A material has Young's modulus E = 200 GPa and experiences a tensile strain of 500 με. Calculate the stress.

12. Explain the working principle of a strain gauge using the resistance equation $R = \rho L/A$. Identify the three quantities that can change.

13. A 120 Ω strain gauge has a gauge factor of 2.1 and is subjected to 500 με strain. Calculate the change in resistance.

14. A strain gauge has an initial resistance of 350 Ω and gauge factor 2.0. If its resistance increases by 0.35 Ω, calculate the strain.

15. A gauge is mounted on a metal structure. The temperature changes even though the mechanical strain is zero. Explain why the measured resistance can still change.

16. State two advantages of using a Wheatstone bridge for strain-gauge measurements.

17. A quarter-bridge contains three 120 Ω resistors and one active strain gauge. The active gauge changes from 120 Ω to 120.24 Ω. The bridge excitation is 6 V. Using the quarter-bridge small-signal relation $V_o \approx (V_{ex}/4)(\Delta R/R)$, calculate the bridge output.

18. Differentiate between uniaxial, biaxial/tri-axial rosette, planar rosette, and stacked rosette selection. Give one appropriate application condition for each.

Instrumentation Amplifiers

19. Why are sensor signals often amplified before being sent to an ADC? State the purpose of buffering and filtering as well.

20. A buffer has high input impedance and its output follows its input. Explain how this prevents the measurement circuit from loading the sensor.

21. An amplifier has a gain of 25 and an input signal of 8 mV. Calculate the output voltage.

22. Define a common-mode signal and explain the purpose of common-mode rejection in an instrumentation amplifier.

23. An instrumentation amplifier has differential gain $A_D = 1000$ and common-mode gain $A_{CM} = 0.01$. Calculate CMRR and CMR in dB.

24. For a subtractor with $R_1 = R_3 = 10\ k\Omega$ and $R_2 = R_4 = 100\ k\Omega$, calculate V_{out} when $V_{IN2} = 25$ mV and $V_{IN1} = 5$ mV. Use $V_{out} = (V_{IN2} - V_{IN1})(R_2/R_1)$.

25. For the classic 3-op-amp instrumentation amplifier, $R_1 = R_3 = 10\ k\Omega$, $R_2 = R_4 = 10\ k\Omega$, $R_5 = R_6 = 10\ k\Omega$ and $R_G = 2\ k\Omega$. Calculate the total differential gain using $G = (1 + 2R_5/R_G)(R_2/R_1)$.

26. Using the same circuit as Question 25, if $V_{IN2} = 12$ mV and $V_{IN1} = 4$ mV, calculate V_{out}.

27. If R_G in Question 25 is changed from 2 kΩ to 1 kΩ, calculate the new total gain. State the relationship between R_G and gain.

28. Why are the subtractor resistors normally kept accurately matched in an instrumentation amplifier instead of using them as the main gain-adjustment resistor?

29. A sensor signal contains a 50 Hz mains component. Explain why filtering is useful and state the purpose of the Nyquist criterion in an ADC interface.